

Rishi Lalwani

✉ rishi.123.lalwani@gmail.com ☎ 6262089545 in LinkedIn ○ Github ○ Portfolio

Education

Madhav Institute of Technology and Science
B.Tech in Internet of Things (IoT)
CGPA: 7.9

Gwalior, Madhya Pradesh
2022 – 2026

Technical Skills

Languages: C++, CSS, HTML, JavaScript, TypeScript

Computer Science Fundamentals: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOPS)

Frameworks/Technologies/Databases: Bootstrap, Express.js, MongoDB, PostgreSQL, Redis, Upstash, Node.js, Next.js, Postman, React.js, Tailwind CSS

Developer Tools & Deployment: Git, GitHub, Vercel, Render, REST API, VS Code

Projects

Movie Insights   [live demo](#) —

Next.js (App Router), React 19, TypeScript, Upstash Redis, Google Gemini AI API, OMDb API, YouTube Data API, Grafana Cloud k6, Vercel

- Developed a full-stack movie insights app utilizing **React Suspense** and **Next.js Server Actions** for deferred streaming of trailer data, cast metadata, and AI audience sentiment.
- Architected a 3-tier caching defense (*In-Flight Promise Map*, *Next.js Data Cache*, and *Upstash Redis*) to eliminate duplicate AI calls, cutting cold container latency from **3.8s to 281ms (13.5x speedup)** and reducing AI token costs by **99%**.
- Implemented atomic distributed locking in Redis via **Lua evaluation scripts** to prevent serverless cache stampedes across Vercel Lambdas, with local heuristic fallbacks guaranteeing **0% server downtime** during API rate-limits (429 errors).
- Conducted protocol load tests at **75 Virtual Users (95.3 peak RPS)** using **Grafana Cloud k6**, maintaining a **0% HTTP error rate** and 100% check pass rate under high concurrency.
- Designed a responsive glassmorphic UI featuring genre carousels powered by Upstash Redis and client-side *localStorage* for personalized user history.

Spotify Wrapped   [live demo](#) —

React.js (Vite), Spotify Web API, OAuth 2.0, Express.js, Node.js, Axios, Vercel

- Implemented **Spotify OAuth 2.0 Authorization Code Flow** with secure access & refresh token handling.
- Developed a **React (Vite) frontend** to visualize user data including top artists, tracks, playlists, and audio features.
- Built a **serverless Express API** on **Vercel** to handle authentication, token refresh, and API proxying.
- Integrated **Spotify Web API** with scoped permissions and error-resilient token refresh logic.

CampTrek (Full-Stack Web App)   [live demo](#) —

Node.js, Express, MongoDB, Passport.js, Cloudinary, MapTiler, EJS

- Developed a campground management platform with **CRUD** functionality.
- Implemented secure authentication with **Passport.js** and cluster maps via **MapTiler**.
- Integrated **Cloudinary** for image management and designed a responsive UI with **Bootstrap EJS**.

Achievements & Certifications

- Secretary – Bandish, Music Club of MITS:** Led creative direction, performed live as a drummer, and co-managed 20+ cultural and musical events.
- Coordinator – ISTE Student Chapter, MITS:** Coordinated technical and cultural events with the Management and PR Committee.
- Postman API Fundamentals Student Expert – 07/2025** 